DOCTORAL RESEARCH PROPOSAL

Multi-Agent Large Language Model Framework for Enterprise Decision Support

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Background & Problem Domain

Enterprises now generate vast volumes of structured and unstructured data, and decision makers need timely, reliable, explainable insight — yet AI adoption remains siloed across business functions.

Traditional Decision Support Systems

- Rule-based and static
- Limited contextual reasoning
- Poor adaptability
- Weak collaboration across business functions

Single LLM Limitations

- Hallucinations
- Limited specialization
- No collaborative reasoning
- Difficult to verify outputs
- Limited explainability

Research Gap: No comprehensive enterprise framework exists that coordinates multiple specialized LLM agents for reliable, explainable, and scalable decision support.

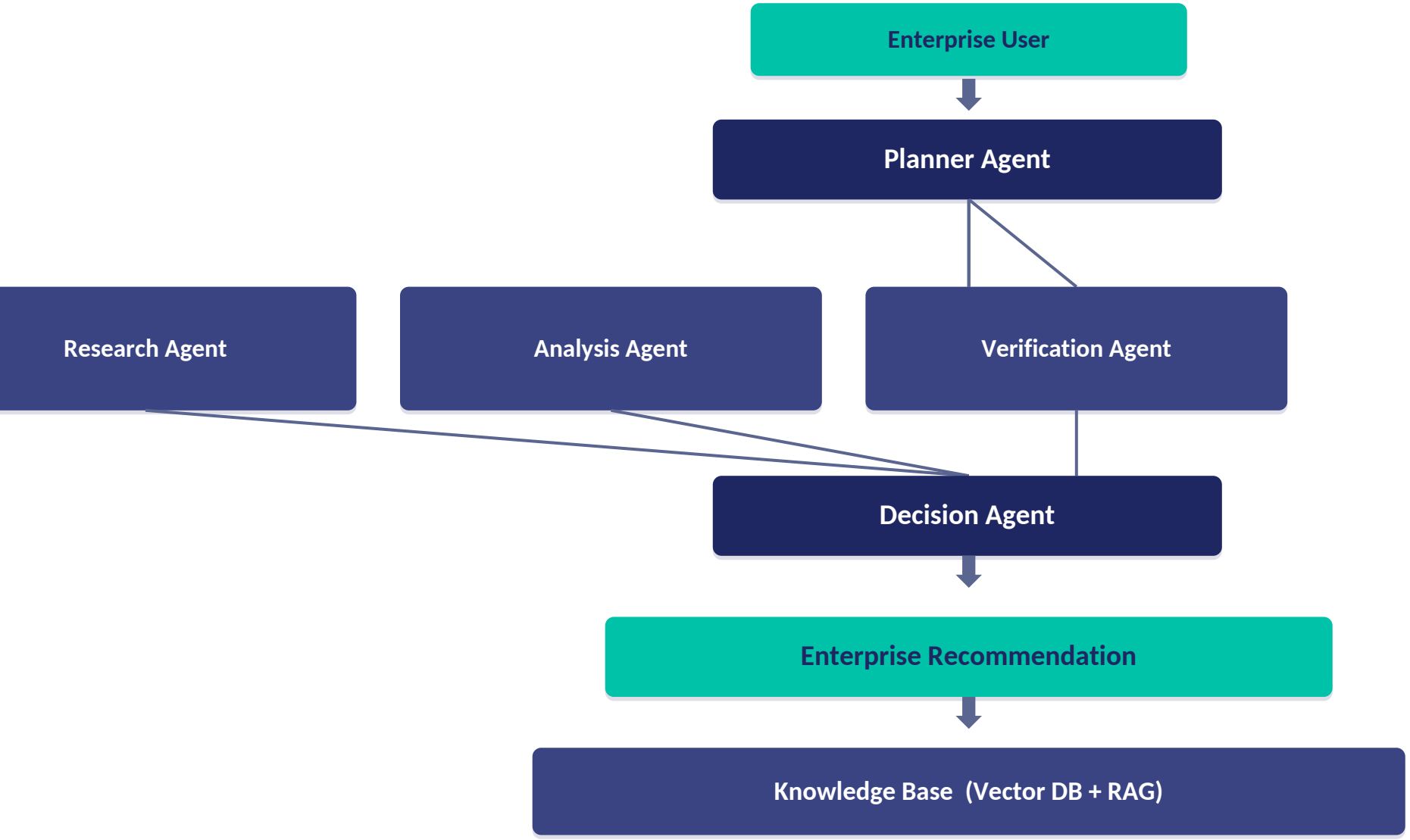
Research Objectives

Primary Objective

Design and develop a Multi-Agent LLM Framework for Enterprise Decision Support that improves decision quality, explainability, and reliability.

- **Design specialized intelligent agents**
- **Enable collaborative reasoning among agents**
- **Integrate Retrieval-Augmented Generation (RAG)**
- **Reduce hallucinations using verification agents**
- **Improve transparency and explainability**
- **Evaluate scalability and enterprise applicability**

Proposed Framework Architecture



TECHNOLOGIES

- GPT / Llama
- LangGraph / CrewAI
- Vector Database
- Knowledge Graphs
- Enterprise APIs

AGENT ROLES

- Planner
- Research
- Analysis
- Verification
- Decision

Research Methodology

1. Literature Review
2. Requirement Analysis
3. Framework Design
4. Multi-Agent Development
5. Knowledge Integration using RAG
6. Experimental Evaluation
7. Comparative Analysis

Evaluation Metrics

- Decision Accuracy
- Precision / Recall
- Hallucination Rate
- Response Time
- Explainability
- User Satisfaction
- Scalability

Expected Outcomes

- Novel Multi-Agent LLM framework
- Higher decision accuracy
- Reduced hallucinations
- Explainable AI recommendations
- Enterprise-ready architecture
- Improved operational efficiency
- Academic publications
- Prototype implementation

Potential Application Domains

- Healthcare
- Finance
- Manufacturing
- Supply Chain
- Smart Governance

Future Scope & Conclusion

FUTURE SCOPE

- ERP and CRM integration
- Autonomous workflow optimization
- Federated Multi-Agent AI
- Human-AI collaborative decision making
- Real-time enterprise analytics

TRIVIA

We had also thought of A Multimodal Agentic AI Framework for Real-Time Carbon Credit Monitoring, Reporting and Verification (MRV)

CONCLUSION

The proposed research addresses the limitations of single LLM systems by introducing a collaborative Multi-Agent framework capable of providing accurate, explainable, and scalable enterprise decision support.

Thank You!